

# Anudeep K

**AWS Cloud Engineer | Cloud Operations | DevOps | Data Engineering**

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## PROFESSIONAL SUMMARY

AWS Certified Solutions Architect - Professional with 9+ years of experience across AWS Cloud, Cloud Operations, Big Data, Data Engineering, ETL, and Enterprise Data Platforms. Currently an AWS Domain Specialist Engineer providing technical expertise for Amazon EMR, Apache Spark, Hadoop, and distributed data workloads, with strong experience in troubleshooting, performance optimization, production support, and Root Cause Analysis (RCA). Hands-on expertise with AWS EC2, S3, EMR, IAM, Lambda, Glue, Athena, CloudWatch, VPC, CloudFormation, Redshift, Auto Scaling, Terraform, CI/CD, Python, Shell, SQL, and Linux, along with Azure Data Factory, Azure Blob Storage, and Azure SQL. Proven track record of improving workload performance, reducing production failures and cloud infrastructure costs, automating operational processes, strengthening security, and improving platform reliability. Experienced in cloud infrastructure, Big Data, ETL, data integration, monitoring, incident management, Infrastructure as Code (IaC), and cross-functional collaboration across enterprise environments.

## TECHNICAL SKILLS

- **Cloud & AWS:** Amazon Web Services (AWS), EC2, S3, EMR, IAM, Lambda, Glue, Athena, CloudWatch, VPC, CloudFormation, Auto Scaling, EventBridge, SNS, SQS, Systems Manager, Secrets Manager, Redshift
- **Cloud & Data - Azure:** Azure Data Factory, Azure Blob Storage, Azure SQL, Cloud Data Migration, Data Integration, Pipeline Monitoring
- **Big Data & Distributed Systems:** Apache Spark, Hadoop, Hive, HDFS, YARN, Kafka, Sqoop, Oozie, Amazon EMR, Distributed Data Processing, Cluster Management, Performance Tuning, Resource Allocation
- **Data Engineering & ETL:** Informatica PowerCenter, ETL Development, ETL Pipelines, Data Integration, Data Migration, Data Validation, Data Cleansing, Data Reconciliation, Data Warehousing, Batch Processing, Workflow Development
- **Programming & Scripting:** Python, SQL, Shell Scripting, Unix/Linux Scripting, Automation, Job Scheduling
- **DevOps & Infrastructure:** Terraform, Infrastructure as Code (IaC), Git, CI/CD, Jenkins, Cloud Infrastructure, Configuration Management, Deployment Automation, Environment Management
- **Databases & Data Platforms:** Oracle, PostgreSQL, SQL Server, Snowflake, Redshift, Azure SQL
- **Monitoring & Operations:** Amazon CloudWatch, Splunk, Linux, Production Support, Incident Management, Root Cause Analysis (RCA), Log Analysis, System Monitoring, Alerting, Capacity Planning, Performance Optimization, SLA Management
- **Security & Reliability:** AWS IAM, Encryption, Network Access Controls, Audit Logging, Security Best Practices, High Availability, Fault Tolerance, Operational Readiness
- **Tools & Methodologies:** Jira, Agile/Scrum, Technical Documentation, Runbooks, SOPs, Knowledge Base Development, Cross-Functional Collaboration

## PROFESSIONAL EXPERIENCE

### AMAZON WEB SERVICES (AWS) - Virginia, USA

#### AWS Domain Specialist | Oct 2025 - Present

- Serve as a domain specialist for AWS Big Data services, providing technical guidance to customers and internal engineering teams to improve workload reliability, scalability, performance, and operational readiness.
- Lead technical investigations for complex customer-impacting issues involving Amazon EMR, Apache Spark, Hadoop, and distributed systems, analyzing logs, metrics, configurations, and workload behavior to drive resolution.
- Perform structured Root Cause Analysis (RCA) for production incidents and implement permanent corrective actions, contributing to a 30% reduction in repeat escalations over the past two quarters.
- Advise customers on EMR cluster configuration, Spark performance optimization, workload scaling, resource allocation, and architecture best practices for large-scale data processing environments.
- Evaluate customer architectures and workload patterns and recommend AWS best practices for resiliency, fault tolerance, scalability, security, and operational readiness.
- Author technical documentation, troubleshooting guides, runbooks, and knowledge-base articles adopted across support teams to standardize incident resolution and operational practices.
- Partner with global AWS engineering, infrastructure, and security teams to resolve critical production issues while maintaining aggressive SLA and customer satisfaction targets.
- Identify recurring technical issues and contribute to process improvements, troubleshooting methodologies, and operational playbooks.

#### AWS Big Data Cloud Support Engineer | Sep 2021 - Sep 2025

- Managed and supported enterprise Amazon EMR clusters running Apache Spark, Hive, Hadoop, HDFS, and YARN, handling an average of 25-30 production cases per month.
- Reduced recurring EMR/Spark execution failures by 20% through systematic RCA, performance analysis, and implementation of permanent corrective solutions.
- Tuned Spark executor memory, partitioning, shuffle configuration, and resource allocation, improving job runtime by approximately 25% for high-volume workloads.

- Reduced AWS infrastructure costs by approximately 15% through EMR auto-scaling, cluster rightsizing, capacity planning, and storage optimization.
- Automated recurring operational and monitoring activities using Python and Shell scripting, reducing manual intervention time by 30%.
- Configured Amazon CloudWatch alerts and Splunk dashboards, reducing average incident detection time by approximately 25%.
- Implemented and supported AWS IAM policies, encryption, and network access controls to strengthen security across customer cloud environments.
- Supported Terraform Infrastructure as Code (IaC) configurations for AWS resource provisioning, configuration management, and environment changes.
- Supported CI/CD pipelines for automated deployments and promotion of application and infrastructure changes across environments.
- Managed EMR cluster scaling, storage optimization, workload capacity planning, and production environment operations.
- Collaborated with global engineering, infrastructure, and security teams to resolve critical production incidents under demanding service-level objectives.

## **CIGNA HEALTHSPRING - Nashville, TN**

### **Informatica Developer / Data Engineer | Jul 2018 - Sep 2021**

- Developed and maintained enterprise-scale ETL workflows using Informatica PowerCenter for high-volume data processing across Oracle, SQL Server, flat files, and APIs.
- Designed Informatica mappings, workflows, reusable transformations, data validation, cleansing, and reconciliation processes to improve downstream data quality.
- Tuned Informatica mappings and workflows, reducing batch execution time by approximately 40% for critical nightly data loads.
- Supported enterprise data integration and migration initiatives using Azure Data Factory, Azure Blob Storage, and Azure SQL.
- Scheduled, monitored, and troubleshooted Azure Data Factory pipelines, resolving data-load failures and supporting reliable on-time data delivery.
- Developed Unix Shell scripts for workflow automation, job scheduling, monitoring, and production ETL support.
- Implemented restartability, exception handling, audit logging, and error-management mechanisms to improve ETL reliability and traceability.
- Supported production ETL operations, troubleshooting workflow failures, data mismatches, and data-processing issues across multiple environments.
- Collaborated with business analysts and QA teams on requirements analysis, validation, testing, and deployments across Dev, QA, UAT, and Production environments.
- Participated in Agile ceremonies, sprint planning, release activities, and production support.

## **XTRACT INC - North Carolina**

### **Software Engineer | Nov 2017 - Jun 2018**

- Supported backend applications and enterprise data-processing systems for business-critical environments, including SQL scripting and reporting automation.
- Troubleshooted production incidents and application failures through log analysis, system monitoring, and Linux-based application support.
- Developed Shell scripts to automate repetitive support and monitoring activities and contributed to code reviews and production bug fixes.

## **MICROINFO - New Jersey**

### **System Engineer | Aug 2016 - Oct 2017**

- Monitored enterprise applications and Linux servers and supported batch-processing environments to maintain operational stability.
- Managed job scheduling, file transfers, workflow execution monitoring, and production support activities.
- Participated in on-call rotations for high-priority production incidents and supported timely issue resolution.
- Maintained SOPs and troubleshooting documentation and contributed to process improvements that reduced operational overhead.

## **EDUCATION**

Master's Degree in Information Technology

## **CERTIFICATIONS**

AWS Certified Solutions Architect - Professional